

TULANCINGO, Mexico

WMO: 766340

Lat: 20.0842N

Lon: 98.3575W

Elev: 2214

StdP: 77.41

Time Zone: -6.00 (MEX)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.8	(c) 4.1	(d) -7.2	(e) 2.7	(f) 14.3	(g) -4.8	(h) 3.3	(i) 15.3	(j) 8.6	(k) 17.9	(l) 7.5	(m) 17.9	(n) 0.7	(o) 150	(p) 0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.6	28.8	14.7	27.4	14.4	26.2	14.3	18.1	23.8	17.5	23.1	17.0	22.5	3.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	15.2	20.5	15.7	14.6	19.7	15.1	14.1	18.9	60.9	24.0	58.4	23.2	56.5	22.5	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.7	(b) 7.3	(c) 6.0	(e) -0.8	(f) 30.8	(g) 2.0	(h) 1.0	(i) -2.2	(j) 31.5	(k) -3.4	(l) 32.1	(m) -4.5	(n) 32.6	(o) -5.9	(p) 33.3		
(a) 8.7	(b) 7.3	(c) 6.0	(e) -2.1	(f) 20.4	(g) 1.7	(h) 1.2	(i) -3.3	(j) 21.3	(k) -4.3	(l) 22.0	(m) -5.2	(n) 22.7	(o) -6.4	(p) 23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	13.3	14.9	16.1	18.1	18.7	18.0	17.0	17.0	16.6	15.3	14.3	13.5
	DBStd	2.88	2.53	2.97	3.22	2.76	2.33	1.52	1.23	1.31	1.73	2.32	2.52	2.00
	HDD10.0	16	5	3	2	0	0	0	0	0	0	1	3	3
	HDD18.3	934	157	101	81	38	23	23	44	46	56	94	121	151
	CDD10.0	2233	107	139	192	243	271	241	218	216	197	166	132	111
	CDD18.3	110	0	4	12	31	36	14	3	4	3	2	1	0
	CDH23.3	1333	21	99	209	346	372	131	33	33	19	32	23	15
	CDH26.7	196	1	7	31	68	73	13	1	0	0	0	1	0
Wind	WSAvg	1.8	1.7	1.9	2.0	2.0	1.9	1.9	2.0	2.0	1.8	1.7	1.7	1.5
Precipitation	PrecAvg	389	7	12	12	23	23	45	45	43	60	45	12	7
	PrecMax	814	58	54	81	81	100	152	140	129	306	216	42	26
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	285	12	14	20	20	28	48	45	44	75	55	14	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	27.8	29.2	30.7	30.2	28.1	25.7	25.7	25.2	25.9	25.3	25.0
		MCWB	11.6	12.8	14.4	14.2	15.4	15.6	15.3	16.3	15.4	15.1	14.6	12.6
	2%	DB	23.8	25.9	27.5	28.7	28.8	26.3	24.3	24.2	23.8	24.2	24.0	23.2
		MCWB	11.5	12.2	13.8	14.2	15.3	15.9	15.5	15.7	15.7	15.0	14.2	12.4
	5%	DB	22.4	24.5	25.9	27.1	27.1	24.9	23.1	23.1	22.4	22.9	22.6	22.2
		MCWB	11.3	12.1	13.2	14.1	14.9	15.7	15.5	15.7	15.7	14.5	13.7	12.4
	10%	DB	20.9	22.9	24.1	25.5	25.5	23.3	21.9	22.0	21.1	21.2	20.9	21.0
		MCWB	10.8	12.0	12.8	13.9	14.9	15.6	15.3	15.5	15.4	14.3	13.2	12.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	16.3	17.2	17.6	18.9	18.9	17.9	18.7	18.2	17.8	17.1	15.6
		MCDB	21.1	24.3	24.9	24.9	25.7	24.7	23.0	23.7	22.6	22.9	23.3	21.4
	2%	WB	13.4	14.8	15.7	16.5	17.6	17.7	17.1	17.7	17.3	16.7	15.8	14.2
		MCDB	20.8	22.3	23.9	24.0	24.7	23.4	22.4	22.8	21.7	21.7	21.8	20.7
	5%	WB	12.5	13.9	14.8	15.6	16.7	17.0	16.5	16.9	16.7	15.9	14.9	13.2
		MCDB	19.7	21.2	22.8	23.3	23.9	22.5	21.3	21.7	20.7	20.5	20.3	20.2
	10%	WB	11.7	13.0	13.8	14.8	16.0	16.3	15.8	16.1	16.0	15.2	13.9	12.4
		MCDB	18.6	20.1	21.5	23.0	22.7	21.0	20.3	20.6	19.6	19.4	18.8	19.3
Mean Daily Temperature Range	5% DB	MDBR	14.7	14.9	14.6	14.2	13.6	11.2	10.5	11.0	9.6	11.3	12.5	15.1
		MCDBR	17.0	17.2	16.8	15.9	15.5	13.9	12.4	12.6	11.8	14.1	15.2	17.1
		MCWBR	8.6	8.2	8.1	7.5	7.4	6.8	5.8	6.2	5.8	7.4	7.9	9.6
	5% WB	MCDBR	15.2	15.1	15.4	14.3	13.8	11.6	10.9	11.3	10.0	11.8	13.6	15.7
		MCWBR	8.9	8.6	8.6	7.9	7.5	6.5	6.1	6.4	5.4	6.7	8.0	9.5
Clear-Sky Solar Irradiance	taub	0.363	0.374	0.391	0.439	0.422	0.425	0.406	0.397	0.355	0.387	0.341	0.350	
	taud	2.285	2.258	2.219	2.106	2.182	2.186	2.269	2.295	2.424	2.285	2.399	2.329	
	Ebn at Noon	909	920	915	870	875	865	882	896	936	894	927	909	
	Edh at Noon	123	134	145	164	151	149	137	134	116	129	109	114	
All-Sky Solar Radiation	RadAvg	4.29	5.04	5.73	6.11	5.95	5.51	5.49	5.49	4.74	4.70	4.39	4.23	
	RadStd	0.35	0.46	0.44	0.37	0.39	0.37	0.47	0.29	0.42	0.34	0.37	0.28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.37	+1.05	-1.75	N/A	N/A	N/A	-2	-127	+165	+3

Nomenclature: See separate page